

# Simpson's Rule

## Key Points

To approximate the integral

$$\int_a^b f(x)dx$$

Split the interval from  $a$  to  $b$  into  $n$  strips each of width  $h = \frac{b-a}{n}$ . Here  $n$  must be **even**. Set

$$x_0 = a, \quad x_i = x_{i-1} + h.$$

Set  $y_i = f(x_i)$ . Then

$$\int_a^b f(x)dx \approx \frac{h}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + \dots + 4y_{n-1} + y_n)$$

## Basic Skills

**Q1.** How many strips are needed on the interval from 0 to 6 if they the following widths

- a)  $h = 1$     b)  $h = 2$     c)  $h = 1.5$

**Q2.** If  $h = 0.75$  and  $x_0 = 3$ , find

- a)  $x_1$     b)  $x_2$     c)  $x_3$     d)  $x_5$

**Q3.** For a function  $f(x) = 2x + 1$  and values  $x_0 = 1, x_1 = 3, x_2 = 5, x_4 = 9$ , find

- a)  $y_0$     b)  $y_1$     c)  $y_2$     d)  $h$     e)  $y_5$

**f)** An approximation for  $\int_1^5 f(x)dx$  using Simpson's rule with 2 strips

## Competency Skills

**Q4.** We want to calculate  $\int_0^2 x \cos^2(x) dx$  using Simpson's rule with 4 strips

a) Fill in the following table

b) Using Simpson's rule approximate  $\int_0^2 x \cos^2 x dx$

c) Determine if Simpson's rule is an under or over estimation of the actual value

**Q5.** Approximate  $\int_0^1 x^3 + 2x^2 + 1 dx$  using Simpson's rule with  $h = 0.5$  and then with  $h = 0.2$ . Then state which provides a better approximation

**Q6.** Approximate  $\int_0^1 e^{x^2} dx$  using Simpson's rule with 5 strips

**Q7.** We know the value of a particular function  $f(x)$  at  $x = 0, 0.1, 0.2, 0.4, 0.6, 0.8, 1$  and we want to find an approximate of  $\int_0^1 f(x)dx$  what is the maximum number of strips we can use

| $i$   | 0 | 1 | 2 | 3 | 4 |
|-------|---|---|---|---|---|
| $x_i$ |   |   |   |   |   |
| $y_i$ |   |   |   |   |   |

## Advanced Skills

**Q8.** Whereas the mid-ordinate rule for integration uses straight lines to estimate  $f(x)$ , Simpson's rule uses parabolas  $y = Ax^2 + Bx + C$ .

a) Find the area under the parabola  $y = Ax^2 + Bx + C$  between  $-h$  and  $h$ , by calculating  $\int_{-h}^h Ax^2 + Bx + C dx$ .

b) Assume  $x_0 = -h, x_1 = 0, x_2 = h$ . Find the unique parabola that goes through the points  $(-h, y_0), (0, y_1)$  and  $(h, y_2)$

c) Find the area under the parabola in b) between  $-h$  and  $h$ , by using a)

d) An estimate of  $\int_{x_0}^{x_2} f(x) dx$  is given by c). What will an estimate of  $\int_{x_2}^{x_4} f(x) dx$  be? Note that translating along to x-axis doesn't change area so you can assume  $x_2 = -h$  now.

e) Generalise d) to an estimate of  $\int_{x_i}^{x_{i+2}} f(x) dx$  where  $i$  is even and  $i \leq n - 2$ .

f) By summing over all these estimates deduce Simpson's Rule.

## Extension

**Q9.** The speed of a car is tracked at 6 second intervals for one minute. The recorded speeds were as follows: 0, 4, 7, 10, 11, 12, 12, 12, 10, 5, 0. Estimate the distance that the car travelled.

Hint: How are speed and distance related?